

Google Play Store App Insights (PowerBI)

Case Study Report

Executive Summary

Project Overview

The Google Play Store contains thousands of applications across multiple categories. Understanding app performance, ratings, installs, pricing strategies, content suitability, and update patterns is crucial for developers, marketers, and business stakeholders.

This project aims to transform raw Google Play Store data into an interactive Power BI dashboard that provides meaningful insights about app performance and market trends.

Objectives

- Analyze application ratings and popularity.
- Identify high-performing categories.
- Study install behavior across app segments.
- Compare free and paid applications.
- Understand pricing and size distribution.
- Analyze update trends and content ratings.
- Build an interactive dashboard for decision-making.

Dashboard Pages

1. Overview Dashboard
 2. Apps Rating Analysis
 3. Installs and Popularity Analysis
 4. Pricing & Size Analysis
 5. Update Trends & Content Analysis
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Problem Statement & Business Questions

Problem Statement

The Google Play Store hosts thousands of applications, making it difficult to identify successful app categories, understand user preferences, and evaluate market trends.

Organizations require a centralized analytics solution that provides insights into:

- App popularity
- User ratings
- Install growth
- Pricing strategies
- Category performance
- Content suitability

Business Questions

Ratings Analysis

- Which categories have the highest ratings?
- Which apps are rated above 4.0?
- What content ratings receive better user feedback?

Popularity Analysis

- Which categories generate the highest installs?
- What are the most installed apps?
- How do reviews influence installs?

Pricing Analysis

- What percentage of apps are free vs paid?
- Which categories contain the most paid apps?
- Does pricing impact ratings?

Content Analysis

- Which content rating is most common?
 - How frequently are apps updated?
 - Which genres dominate the Play Store?
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Dataset Understanding

Dataset Description

The dataset contains information regarding Android applications available on the Google Play Store.

Important Attributes

Column Name	Description
App	Application Name
Category	App Category
Rating	User Rating
Reviews	Number of User Reviews
Size	App Size
Installs	Total Installs
Type	Free/Paid
Price	App Price
Content Rating	Target Audience
Genres	App Genre
Last Updated	Last Update Date
Android Ver	Minimum Android Version

Dataset Characteristics

- Multiple app categories
 - Free and paid applications
 - Various content ratings
 - Historical update information
 - User review and install metrics
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Data Cleaning & Preparation

Data Quality Issues Identified

Several columns contained inconsistent formats and missing values.

Cleaning Activities Performed

1. Size Column Transformation

Issues:

- Values stored in MB and KB format.
- Text values prevented numerical calculations.

Actions:

- Converted KB values into MB.
- Removed text suffixes.
- Converted column into decimal format.

Example

Original Converted

512KB 0.5 MB

25M 25 MB

2. Installs Column

Issues:

- Contained symbols such as "+" and commas.

Actions:

- Removed special characters.
 - Converted to whole number datatype.
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3. Price Column

Issues:

- Included "\$" symbols.

Actions:

- Removed currency symbols.
 - Converted to decimal format.
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4. Last Updated Column

Actions:

- Converted to Date datatype.
 - Extracted Year for trend analysis.
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5. Missing Values

Actions:

- Removed blank and invalid records.
 - Validated rating and review fields.
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Dashboard Design & Overview Analysis

Overview

Purpose:

Provide a high-level summary of Play Store performance.

Key Performance Indicators

- Total Installs
- Average Rating
- Total Reviews
- Total Apps

Visuals Used

Apps Success Matrix

Analyzes:

- Category Performance
- Rating Distribution
- Install Distribution

Apps Distribution

Displays:

- Free vs Paid applications.

Performance Breakdown Table

Provides:

- App-level performance metrics.

Key Insight

The Play Store ecosystem is dominated by a few categories with significantly higher installs and ratings than others.

Apps Rating Analysis

KPIs

- Average Rating
- Apps Rated 4+
- Highest Rated Category
- Highest Rating

Key Findings

Category Performance

Certain categories consistently achieve higher user satisfaction scores.

Rating Distribution

Most applications fall within the 4–5 rating range, indicating generally positive user feedback.

Content Rating Analysis

Apps targeted toward general audiences tend to dominate the platform.

Top Rated Applications

Several applications maintain ratings above 4.5 while accumulating significant review counts.

Installs & Popularity Analysis

KPIs

- Total Installs
- Average Installs
- Most Popular Category
- Highest Installed App

Key Findings

Category Popularity

Game applications receive the largest number of installs.

Reviews vs Installs

A positive relationship exists between user reviews and install counts.

App Type Distribution

Free applications dominate total installs.

Top Installed Apps

A small number of applications account for a significant portion of overall downloads.

Pricing & Size Analysis

Objective

Understand how pricing and application size affect app performance.

KPIs

- Average Price
- Paid Apps
- Free Apps
- Most Expensive App

Analysis Performed

Price Distribution

Apps were grouped into pricing brackets:

- Free
- \$0–1
- \$1–5
- \$5–10
- \$10–50
- Above \$50

App Size Analysis

Applications were categorized based on size ranges.

Price vs Rating

Evaluated whether expensive applications receive higher ratings.

Size vs Installs

Studied the impact of application size on popularity.

Key Findings

- Free apps dominate the marketplace.
 - Higher prices do not guarantee higher ratings.
 - Medium-sized applications often achieve stronger adoption.
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Update Trends & Content Analysis

KPIs

- Updated Apps
- Latest Update Year
- Most Common Content Rating
- Total Genres

Key Findings

Update Trends

Application updates increased significantly over time, reflecting continuous developer engagement.

Content Rating Distribution

The majority of applications are designed for general audiences.

Genre Analysis

Certain genres consistently contain a higher number of applications.

Recently Updated Apps

Recently maintained applications generally demonstrate stronger user engagement.

Conclusion

The Google Play Store ecosystem is highly competitive and heavily influenced by user ratings, install volume, content suitability, and continuous updates.

Key observations include:

- Free applications dominate the platform.
 - Games represent the most popular category.
 - Ratings above 4.0 are common among successful apps.
 - Regular updates contribute to app relevance.
 - User reviews strongly correlate with popularity.
 - Content targeting broad audiences achieves greater reach.
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Recommendations

For Developers

- Focus on maintaining ratings above 4.0.
- Release frequent updates.
- Monitor user feedback regularly.
- Optimize app size for better adoption.

For Business Stakeholders

- Prioritize high-growth categories.
- Invest in applications with strong review performance.
- Track update frequency as a success indicator.

Future Enhancements

- Predictive install forecasting.
- Sentiment analysis on reviews.
- Category-level revenue modeling.
- Machine learning-based app success prediction.